# LUMORA: COGNITIVE-STYLE AND COGNITIVE-LOAD-AWARE ADAPTIVE LEARNING

INDIVIDUAL THESIS

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Individual component: Resource Upload and Source-Grounded Adaptive Content Conversion

Group ID: R26-SE-004

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science Honours in Information Technology

Specialisation in Software Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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# ABSTRACT

Adaptive learning requires a reliable connection between educator-authored resources and learner-specific explanations. In Lumora, that connection is the individual resource upload and content conversion component, which processes lecture videos, PowerPoint presentations, and PDF documents into lesson knowledge used for cognitive-style and cognitive-load-aware adaptation. This thesis presents an implemented pipeline combining asynchronous subsection processing, speech transcription, document extraction, mathematical-content handling, semantic near-duplicate removal, and dual-model explanation selection. Uploaded assets are stored before processing is scheduled through a single-process first-in, first-out queue. The knowledge assembly stage retains extracted source wording, prioritises presentation material, and adds complementary document and transcript content. Mathematical resources follow dedicated Office Math Markup Language and PDF extraction paths, with canonical LaTeX equations supplied to the generation prompt. DeepSeek and Hugging Face candidates are ranked using a weighted combination of source similarity and cognitive-load-targeted readability. Repository inspection and reproducible offline verification were used to establish implementation behavior. Eighteen of nineteen focused checks passed; a PDF fixture failed a symbol-preservation check despite producing non-empty extracted equations. The PowerPoint fixture returned five equations, and the output selector preferred the source-aligned candidate in a controlled lexical-fallback example. These findings demonstrate selected technical capabilities and expose limitations in equation fidelity, readiness criteria, similarity-based faithfulness estimation, and degraded operation. They do not establish learning improvement or a measured reduction in hallucination. The individual contribution is an integrated resource-to-adaptation workflow with inspectable selection metadata and documented failure modes. Further work should introduce teacher-reviewed extraction, stricter quality gates, calibrated factuality evaluation, and controlled learner validation.



Keywords: adaptive content conversion, resource processing, mathematical extraction, microlearning, hallucination mitigation

# ACKNOWLEDGEMENTS

I acknowledge Dr Kalpani Manathunga, supervisor, and Mr Jeewaka Perera, co-supervisor, for their supervision of the Lumora research project. I also acknowledge the collaboration of Dilshan W.G. W, Imesh L.A.R, and Subasinghe M.H.R.I, whose explainable AI, cognitive style detection, and cognitive load detection components provide the wider context for the content conversion workflow. The Sri Lanka Institute of Information Technology is acknowledged as the institution within which this individual thesis is submitted. No funding source or additional technical assistance is asserted because such details were not supplied.

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# LIST OF ABBREVIATIONS

| Abbreviation | Meaning |
|---|---|
| AI | Artificial intelligence |
| API | Application programming interface |
| ASR | Automatic speech recognition |
| FIFO | First-in, first-out |
| FRE | Flesch Reading Ease |
| HF | Hugging Face |
| HTTP | Hypertext Transfer Protocol |
| JWT | JSON Web Token |
| LLM | Large language model |
| MCQ | Multiple-choice question |
| OMML | Office Math Markup Language |
| PDF | Portable Document Format |
| PPTX | Office Open XML presentation format |
| RAG | Retrieval-augmented generation |
| SBERT | Sentence-BERT |
| SDG | Sustainable Development Goal |
| TF | Term frequency |
| XML | Extensible Markup Language |

# 1 INTRODUCTION

## 1.1 Overall project context

Lumora is a multi-component adaptive learning platform intended to connect educational content with a learner's cognitive characteristics and changing learning effort. Teachers create courses and upload lecture materials. Students interact with video lessons and receive generated learning aids. Cognitive style detection uses webcam and cursor behavioral information to estimate presentation-related and analytic–wholistic characteristics. Cognitive load detection uses video interaction features, including pausing and rewinding, to estimate learning effort. Explainable artificial intelligence (AI) presents model-related insights to teachers, while course-linked exam aids provide practice questions from educator resources.

The full platform therefore requires more than prediction models. It needs a content infrastructure that can turn a teacher's heterogeneous files into usable evidence and then pass that evidence to the adaptation process. Without this infrastructure, a model may generate a readable explanation that does not represent the lecture, or it may receive duplicated and damaged material that compromises the adaptation. The individual component addressed in this thesis supplies this resource-to-explanation connection.

The complete application uses a React interface, an Express application programming interface (API) gateway, and separate Node.js and Python services. Course and resource information are represented in MongoDB. Other components use their own services and databases. This thesis discusses those neighboring components only where they provide inputs or receive outputs; it does not claim ownership of their cognitive prediction or explanation algorithms.

## 1.2 Individual component context

The individual component includes teacher-side resource upload, background preparation of lesson knowledge, and adaptive content conversion. A course contains ordered sections and subsections. Each newly created subsection requires a video and can include a PowerPoint file, a PDF, and images. Resources are stored externally, and the subsection record holds their URLs and processing metadata. Subsequent processing creates

transcript text, document extracts, extracted images, deduplicated knowledge, and canonical equation entries where mathematical handling is enabled.

The adaptation stage combines that knowledge with a learner profile and cognitive state. The inspected implementation builds an instructional prompt and requests candidates from DeepSeek and a Hugging Face-hosted model. It then selects an output using a source-similarity proxy and readability alignment. The term dual conversion in this thesis refers to two candidate transformations of the same instructional input. It does not refer to a voting ensemble, a trained multimodal fusion model, or agreement between two independently verified factuality judges.

The candidate is Jayasinghe J. M. S. N, student ID IT22166074. The scope of contribution is based on the responsibility supplied by the candidate and the associated code paths. Repository inspection establishes implementation behavior; it does not independently establish personal authorship of every line. Individual contribution should be confirmed against the team's contribution records before submission.

## 1.3 Problem statement

Educator-authored lecture resources often contain overlapping information across spoken explanations, slides, and supplementary documents. Ordinary concatenation can increase prompt length while repeating the same facts. Plain text extraction can damage equations, omit images, and remove structural information needed to interpret a lesson. Processing these resources synchronously can also keep a teacher waiting while transcription and extraction complete.

A further problem arises during generative conversion. A large language model (LLM) may produce fluent instructional text that changes a mathematical expression or introduces a claim absent from the educator's resources. Maynez et al. document faithfulness problems in abstractive summarization [1]. Their findings motivate explicit grounding and validation, but do not prove that any particular prompt or similarity score eliminates such errors. The problem addressed here is therefore how to prepare manageable lesson evidence and transform its presentation while maintaining a defensible relationship to that evidence.

## 1.4 Research gap

The reviewed literature provides techniques for sentence embeddings, mathematical recognition, microlearning, and generated-content evaluation. These techniques address different parts of the problem. Sentence embeddings support semantic comparison [2], formula recognition research highlights the difficulty of reconstructing mathematical structure [5], [6], and microlearning research investigates focused instructional units [9], [10]. None of these references alone establishes a complete implementation for Lumora's combination of upload processing, equation protection, style-aware prompting, and two-candidate selection.

The proposed gap is an integration and evaluation gap: establishing how these mechanisms can be connected within a practical educator-to-learner workflow, and where their combined reliability breaks down. It is not a claim that no prior adaptive learning system exists. The supplied spreadsheet is a seed bibliography rather than an exhaustive review of all competing platforms. Consequently, the thesis frames originality at the level of the component design and its investigated trade-offs, subject to a broader literature review before final submission.

## 1.5 Aim and objectives

The aim is to develop and assess a resource upload and adaptive conversion component that produces source-grounded learning aids from heterogeneous lecture materials while supporting manageable instructional units and inspectable output selection.

The objectives are to: (O1) accept and organise teacher resources with controlled access and bounded uploads; (O2) prepare subsection knowledge asynchronously with observable processing states and restart recovery; (O3) extract text, transcripts, images, and mathematical representations through appropriate processing paths; (O4) reduce repeated lesson information using semantic or lexical comparison without introducing a generative summarizer into knowledge assembly; (O5) transform lesson presentation using cognitive style and load inputs; and (O6) compare two generated candidates using source similarity and readability, while exposing limitations and degraded modes.

## 1.6 Research questions or hypotheses

RQ1 asks whether the resource preparation workflow produces usable, traceable lesson knowledge from presentation, PDF, and video inputs. RQ2 asks how source-prioritised deduplication affects redundancy, complementary information retention, and equation preservation. RQ3 asks whether the two-candidate selector improves human-rated source consistency and presentation suitability compared with either candidate alone. RQ4 asks how background processing and fallback mechanisms behave when extraction, embedding services, or generation services are unavailable.

RQ1 and selected parts of RQ4 are examined through implementation inspection and focused offline tests. RQ2 requires a labeled content-retention benchmark beyond the small fixtures used here. RQ3 requires generated samples and blinded expert ratings; no outcome is asserted for it. A testable future hypothesis is that two-candidate selection improves source-supported claim precision relative to an unselected single-model baseline, without reducing concept coverage below an agreed threshold. This remains a hypothesis, not a reported finding.

## 1.7 Scope, limitations and assumptions

The component operates on teacher-uploaded materials associated with an identified course subsection. It assumes that the teacher is authorised to use those materials and that learner-state inputs are supplied by the other components. New subsections require video, while document and image attachments are optional. The primary presentation parser supports PPTX; legacy binary PPT uploads are accepted by the upload filter but do not produce text through the inspected parser. Text-layer PDF extraction is implemented, but general scanned-document recognition is not demonstrated.

The current implementation restricts uploaded videos to fifteen minutes. This is an engineering limit supporting shorter instructional units, not a validated optimal microlearning duration. Prompt content is bounded by a character budget rather than a measured token budget. Source similarity is a proxy for grounding and must not be interpreted as factual entailment. Learner studies, model accuracy results, production latency measurements, and independent teacher validation were not supplied and were not manufactured for this thesis.

## 1.8 Individual contribution

The component's principal contribution is the coordinated preparation and conversion of resources. The workflow connects upload validation, external asset storage, a background worker, format-specific extraction, deduplicated lesson knowledge, math-aware prompts, and candidate selection. It includes operational metadata so that processing and selection decisions can be inspected. Its engineering contribution is strongest when viewed as a linked pipeline, because failures early in extraction affect later transformation even when the generation model behaves correctly.

The thesis additionally contributes a reproducible evidence package prepared for this review: deterministic checks, machine-readable outcomes, a source-code traceability map, and an audit of the supplied bibliography. These checks are newly executed verification evidence, not historical experiment results from the original project development period.

## 1.9 Thesis structure

Chapter 2 reviews relevant work and distinguishes direct evidence from conceptual support. Chapter 3 describes the research approach, verification procedure, and proposed empirical evaluation. Chapter 4 presents requirements and solution design. Chapter 5 explains implementation, integration, and test procedures. Chapter 6 reports observed technical results and missing validation. Chapter 7 interprets the findings and limitations. Chapter 8 concludes with objective achievement and future directions. Appendices provide code traceability, reference auditing, and AI-use disclosure.

# 2 LITERATURE REVIEW AND RELATED WORK

## 2.1 Review method and scope

The review started from the eighteen populated reference entries in Research Papers Evidence.xlsx. References were grouped by mathematical processing, near-duplicate detection, microlearning, similarity thresholds, chunking, output evaluation, and prompt design. Primary paper pages, author-hosted material, publisher records, and the authors' own technical article were inspected where accessible. Bibliographic titles and authors were corrected where the spreadsheet used a proceedings title or an incomplete description. Access limitations and uncertain claims are recorded in Appendix A.

This procedure is a targeted narrative review. It is not a systematic review conducted through predefined database queries, duplicate screening, and formal inclusion decisions. The spreadsheet's summaries are treated as candidate interpretations rather than evidence that a claim is established. Papers from other domains are used cautiously: their methods may motivate a design, but their thresholds and results do not automatically transfer to instructional content.

## 2.2 Domain and theoretical foundations

### 2.2.1 Source grounding and faithfulness

Source faithfulness concerns whether generated information is supported by the supplied material, whereas factuality may also consider correctness against knowledge outside that material. A generated statement can be generally true but unsupported by the uploaded lesson, or it can closely resemble a mistaken source. Maynez et al. show why fluent abstractive output requires more careful assessment than surface quality alone [1]. In this component, extractive knowledge assembly avoids adding another generative rewriting stage before adaptation. It does not remove transcription errors, extraction errors, incorrect teacher material, or hallucination in the adaptation stage itself.

### 2.2.2 Cognitive load and presentation

Sweller's cognitive load theory relates instructional processes to limited cognitive processing capacity and schema acquisition [11]. It provides a rationale for considering unnecessary complexity in learning support. This rationale motivates concise

explanations and a readability target informed by the supplied load estimate. However, Flesch Reading Ease (FRE) measures features of English text, not the learner's cognitive load. A load-aware FRE target is an implementation hypothesis that needs empirical calibration, especially for technical vocabulary and mathematical content.

Felder and Silverman discuss learning and teaching preferences in engineering education [12]. Their work supplies background for varied presentation, but does not validate Lumora's webcam-based classification or prove that matching a detected category improves learning. Cognitive style, learning preference, and learning outcome must remain distinct constructs. The conversion component receives classifications from adjacent services and transforms layout; validating those classifications lies outside this individual component's scope.

### 2.2.3 Microlearning and knowledge chunking

Microlearning is an instructional organisation strategy, while technical chunking is a processing operation. Monib, Qazi, and Apong review microlearning across contexts and propose a framework for its design [9]. Stavrinou et al. evaluate a short-form educational video system and report favorable outcomes in their own study [10]. These works motivate focused, navigable learning units. Their findings are not Lumora results and do not establish the fifteen-minute upload ceiling as educationally optimal.

The implementation includes subsection-level organisation, thirty-second transcription buckets, sentence-like deduplication units, and a prompt character ceiling. Each operates at a different scale. A thirty-second ASR bucket is not necessarily a coherent learning object. Sentence splitting does not guarantee that a definition and its exceptions remain together. Meaningful microlearning requires objectives and concept boundaries in addition to technical length constraints.

## 2.3 Related research

### 2.3.1 Semantic representations and redundancy

Sentence-BERT creates sentence representations suitable for semantic comparison [2]. MiniLM investigates distillation of transformer attention to obtain compact representations [3]. These studies motivate the use of all-MiniLM-L6-v2 through the sentence-transformers package in the deduplication stage. RETSim investigates robust

near-duplicate retrieval and deduplication [4]; it is relevant as an alternative representation approach, but RETSim is not implemented in the inspected pipeline.

Cosine similarity enables comparison between chunk vectors. It does not by itself distinguish a useful paraphrase from a repeated fact, nor does it prove that a chunk has no unique detail. Hanley and Durumeric's information-flow study [13] and Ai et al.'s document novelty metric [14] illustrate similarity and novelty analysis outside education. They do not validate Lumora's particular source-specific cutoffs. The thresholds in this component must therefore be described as configured heuristics pending labeled calibration.

### 2.3.2 Mathematical content conversion

Gebhard and Rosenblum describe the challenges of translating different Word equation representations into publication formats [5]. Jiang et al.'s LATTE addresses recognition and iterative refinement of formulas and tables [6]. Horn and Keuper provide a mathematical extraction benchmark using formula ground truth and meaning-sensitive evaluation [7]. Together these sources support treating equations as structured content rather than ordinary words. LATTE and the benchmark evaluation method are related work; neither is claimed as an implemented subsystem.

Jung et al.'s MathBridge addresses the relationship between spoken mathematical expressions and LaTeX representations [8]. It helps identify an important distinction: transcribing mathematical speech is not equivalent to recovering a faithful equation. Lumora stores ASR output and separately extracts equations from documents. It does not implement or fine-tune MathBridge. Delimiter conventions such as dollar-marked expressions are useful representation choices, not evidence that the recovered formula is correct.

### 2.3.3 Chunk boundaries and output assessment

Gomez-Cabello et al. compare chunking strategies in a clinical decision-support setting [15]. Their work motivates preserving coherent evidence boundaries and evaluating segmentation choices, while its domain-specific results cannot be treated as instructional performance estimates. Lumora's own prompt ceiling and sentence splitting should be evaluated independently rather than justified by adopting another study's numerical settings.

Li et al. study readability and faithfulness of concept-based explanations of model behavior [16]. Their constructs differ from Flesch-based readability and source grounding of lecture rewrites; the spreadsheet's proposed direct equivalence is therefore too strong. The Databricks authors' evaluation guidance discusses assessment dimensions for LLM applications [17], but does not establish the exact 0.65/0.35 weights used in this project. Xu et al.'s ExpertPrompting motivates structured expert-role prompting [18]; it does not prove that Lumora's shorter pedagogical role instruction has the same effect.

## 2.4 Existing systems, products and techniques

The component draws on reusable libraries and services rather than proposing a new foundational language model. Whisper performs ASR, Cloudinary stores assets, pdf-parse obtains ordinary PDF text, PyMuPDF supplies geometry-aware extraction, and a custom OMML converter handles embedded presentation equations. Sentence-transformers provides embeddings for deduplication. DeepSeek and Hugging Face provide generative candidates, while a custom selector ranks them. These names identify implementation dependencies, not an experimentally established product comparison.

A conventional resource repository can store and display files without converting them into learner-specific explanations. A single-model adaptation workflow can rewrite material without offering candidate-level comparison. A general RAG system usually retrieves indexed material in response to a query; Lumora's core lesson conversion instead assembles known subsection evidence and passes it to generation. Accordingly, the implemented architecture is best described as subsection-grounded conversion with retrieval-like preprocessing, rather than a demonstrated global vector-database retrieval system.

Table 2.1 summarises the relationship between literature and design. Exact operational settings remain component decisions.

Table 2.1. Literature-to-design synthesis.

| Theme | References | Supported design rationale | Remaining evidence requirement |
|---|---|---|---|
| Source consistency | [1] | Separate extracted evidence from generated adaptation | Human judgment of supported and unsupported claims |
| Semantic comparison | [2]–[4] | Compare paraphrased instructional chunks | Calibrate deletion thresholds and unique-detail retention |

| Theme | References | Supported design rationale | Remaining evidence requirement |
|---|---|---|---|
| Structured mathematics | [5]–[8] | Use specialised equation handling | Compare recovered expressions with verified originals |
| Focused learning units | [9]–[11] | Organise concise lesson support | Test learning outcomes and coherent concept boundaries |
| Preference-aware presentation | [12] | Offer different explanatory layouts | Validate classification and adaptation utility |
| Similarity and novelty | [13], [14] | Investigate selective redundancy reduction | Avoid importing unrelated numerical thresholds |
| Chunk boundary quality | [15] | Preserve evidence context | Evaluate local sentence and prompt truncation policies |
| Evaluation and prompting | [16]–[18] | Inspect multiple quality dimensions | Calibrate metrics, weights, and role instructions |

## 2.5 Gap synthesis and conceptual framework

The conceptual framework has three linked responsibilities: evidence preparation, controlled presentation transformation, and inspectable selection. Evidence preparation should retain the meaning and provenance of teacher material. Transformation should respond to learner-state inputs while staying within the source boundary. Selection should prefer an appropriate candidate and disclose whether its scoring method has degraded. Each responsibility requires an independent measure; one composite score cannot validate the whole pipeline.

The resulting research proposition is that adaptation reliability depends on upstream resource quality as well as downstream generation quality. Correctly formatted generated mathematics can still be wrong if the canonical extraction is wrong. Likewise, short output can still omit an essential condition. The framework therefore treats content coverage, mathematical fidelity, source-supported claim precision, and operational completion as distinct evaluation targets.

# 3 RESEARCH METHODOLOGY

## 3.1 Research design

The thesis uses an artefact-centered software engineering approach with technical verification. The artefact is the implemented resource processing and conversion pipeline. Its requirements were derived from the supplied project description, and its mechanisms were examined through code tracing. A focused offline test suite was then created and executed against available local modules and fixtures. This establishes selected observable behaviors without representing the tests as a complete experimental evaluation.

A fuller design-science evaluation remains proposed. It should compare alternative resource preparation and candidate-selection strategies using fixed materials, predefined outcome measures, and independent ratings. The current report separates implementation findings from these future empirical questions. It does not retrofit a participant study, sample size, ethics approval, or statistical significance onto code inspection.

## 3.2 Data, participants or study materials

Three evidence sources were available: the current repository, the supplied thesis-content PDF, and the research-evidence spreadsheet. Two existing mathematical fixtures were used: heat_equation_slides.pptx and heat_equation_notes.pdf. These fixtures were generated for upload testing and are not a representative sample of real institutional teaching resources. Short authored text inputs were used for controlled deduplication, prompting, and candidate-selection checks.

No learners, teachers, interview responses, consent forms, or historical generated-output benchmark were provided. Therefore participant counts, demographic characteristics, usability ratings, and learning gains are not reported. If user validation is later undertaken, it should recruit consenting participants according to institutional approval and record the inclusion criteria and study setting.

## 3.3 Experimental or study procedure

The technical procedure comprised five stages. First, relevant controllers, services, model schemas, frontend orchestration, and helper modules were traced. Second, configured

thresholds and limits were extracted from code rather than inferred from spreadsheet statements. Third, references were checked for title, authorship, subject, and relevance. Fourth, deterministic checks were run without database connections or generative-model requests. Fifth, observed behavior and limitations were consolidated into the thesis with file-level traceability.

The JavaScript verification contains thirteen checks. These cover elementary OMML conversion, fixture extraction, legacy PPT behavior, source ordering, quality warnings, prompt layouts, FRE mapping, lexical similarity fallback, two-candidate ranking, single-candidate handling, and empty-candidate handling. The Python verification contains six checks covering exact duplicate removal, math protection, transcript boundaries, PDF extraction availability, transcript sampling, and PDF heat-equation symbol fidelity. Remote embedding access was deliberately disabled so the tests could be reproduced without credentials. Tests of lexical fallback must not be described as tests of neural MiniLM quality.

For the proposed benchmark, collect a varied set of authorised lecture resources spanning text-heavy lessons, mathematical lessons, and mixed visual material. Obtain teacher-verified expected facts and equations. Run raw concatenation, exact-deduplication, and semantic-deduplication variants over the same materials. Generate candidates under fixed profile inputs and record the selected model and scoring mode. Have reviewers assess randomly ordered outputs without seeing model names. Freeze prompts, model identifiers, generation settings, and reference extracts for each run.

## 3.4 Measures and evaluation metrics

Table 3.1 distinguishes metrics actually observed from those requiring additional data. A future benchmark should pair automatic measures with human review. Source-supported claim precision can be defined as the number of supported generated claims divided by all checkable generated claims. Concept coverage can be defined against a teacher-verified concept inventory. Unsupported-claim rate and omission rate should be measured separately because avoiding new claims does not ensure a complete explanation.

Table 3.1. Evaluation measures and evidence status.

| Measure | Operational definition | Evidence available |
|---|---|---|
| Focused verification outcome | Pass/fail for predefined offline assertions | Observed: 18 passes, 1 failure |

| Measure | Operational definition | Evidence available |
|---|---|---|
| Extract availability | Non-empty text and equation list for a fixture | Observed for PPTX and PDF |
| Formula fidelity | Preserved symbols and teacher-verified mathematical meaning | One targeted PDF symbol check failed; broader evaluation pending |
| Duplicate reduction | Dropped chunks divided by evaluated input chunks | Observed only for an exact-duplicate fixture |
| Unique concept retention | Retained verified distinct concepts divided by source concepts | Not measured |
| Selection behavior | Selected model and recorded scores for fixed candidate texts | Observed under lexical fallback |
| Generated claim precision | Source-supported claims divided by checkable claims | Not measured |
| Processing latency | Upload acceptance, queue wait, and preparation time measured separately | Not measured |
| Learner outcome | Quiz gain, delayed retention, and effort ratings | Not measured |

The implemented selection score is $C = 0.65F + 0.35R$, where F is clamped source-to-candidate cosine similarity when available and R is readability-target alignment. $FRE = 206.835 - 1.015(W/S) - 84.6(Y/W)$, where W, S, and Y are the heuristic word, sentence, and syllable counts. $R = 1 - \min(1, |FRE - T|/50)$, with T determined by load level. The current map is 75 for Very High, 65 for High, 55 for Medium, and 45 for Low. Unmapped labels use 55. These are configured values, not empirically fitted physiological or pedagogical thresholds.

## 3.5 Data analysis

For executed checks, analysis is descriptive. Assertions are reported individually with fixture details and methodological limits. A pass confirms its stated predicate, not overall system correctness. In particular, the non-empty PDF check and the symbol-fidelity check answer different questions. Reporting both prevents availability from being misrepresented as mathematical accuracy.

Future paired evaluations should aggregate outcomes at lesson level because multiple outputs from one lesson are not independent observations. Reviewer agreement should be measured before combining ratings. Report distributions, uncertainty intervals, and effect sizes, and select inferential procedures according to the observed data and study design.

Sensitivity analyses should vary similarity thresholds, scoring weights, model availability, and prompt length. No statistical hypothesis test was performed for this draft.

### 3.6 Ethics, privacy, security and responsible AI

Teachers must have permission to upload and transform educational material. Student identity and learner-state data should be limited to what conversion requires. Generation prompts may disclose lesson text to external inference providers; deployment documentation should identify that flow and the applicable institutional data-handling arrangements. The component does not need to claim that raw webcam footage is sent to generation: the inspected prompt consumes profile labels and load inputs, while behavioral collection belongs to another component.

Access controls include JSON Web Token (JWT) verification and educator ownership checks in subsection controllers. Such controls are implementation mechanisms, not evidence of a completed security audit. The inspected authentication helper contains a fallback secret, and some file validation relies on extensions or media types. Production operation should require configured secrets, stronger content inspection, resource quotas, and approved retention and deletion procedures. Extracted canonical equations also need teacher review before they are presented as authoritative.

Generated material should identify its relationship to the original lesson and disclose degraded scoring. Further-reading suggestions are separate from source-constrained adaptation because they intentionally extend beyond original lesson material. Academic integrity obligations apply to this thesis as well: AI drafting is disclosed in Appendix B, signatures remain blank, and the candidate must verify attribution and approve the final text.

### 3.7 Reproducibility and research artefacts

The inspected repository commit is 7a8af53deda48d6b91dc2e17b8fef7bda771dfc7. The working-tree state at the time of review may include local changes and should be archived with the evidence package. Verification was executed in October 2026 using Node.js 22.20.0 and the available Python environment. Verification scripts and their JSON outputs accompany this thesis. They do not load environment files or call external generation APIs.

The artefacts include the thesis source, editable Word document, PDF, reference audit, code traceability information, deterministic test scripts, and recorded outcomes. For final research reproducibility, additionally record package versions, model revisions, random seeds where applicable, hardware, runtime configuration excluding secrets, and the exact authorised teaching corpus. Screenshots and service-level logs remain to be added if available from the original project operation.

# 4 PRODUCT AND SOLUTION DESIGN

## 4.1 Requirements

The workflow must accept educator resources, associate them with course structure, and avoid keeping the upload request open for the complete transcription and extraction workload. It must produce usable lesson evidence with special handling for mathematical content, preserve source association, and pass bounded knowledge to adaptation. It must also provide teacher-visible processing information and identify when candidate selection uses a fallback.

Table 4.1 lists implemented mechanisms and their limits. Functional implementation is distinct from verified deployment.

Table 4.1. Requirements and implemented mechanisms.

| Requirement | Mechanism | Qualification |
|---|---|---|
| Authorised upload | Gateway checks, JWT verification, educator ownership validation | Security penetration testing not supplied |
| Bounded resource input | File size checks and fifteen-minute video validation | Does not enforce instructional coherence |
| Responsive teacher workflow | Asset storage followed by HTTP 202 and scheduling | External upload completes before acceptance response |
| Background preparation | Global FIFO and child-process extraction | Single Node process, not a distributed queue |
| Recover interrupted jobs | Startup scan of queued/processing subsection records | No cross-instance lease or preserved original FIFO order |
| Preserve mathematical material | Math flag, OMML/PDF paths, canonical LaTeX entries | Extraction fidelity remains format-dependent |
| Reduce duplication | Source-specific cosine thresholds and math protection | Unique details can be incorrectly removed |
| Adapt content | Profile- and load-conditioned prompt | Prompt rules are not hard guarantees |
| Compare candidates | Weighted source similarity and readability | Relative ranking without a factuality acceptance gate |

## 4.2 Component architecture within the complete solution

Figure 4.1 shows the main dependency chain. Teachers interact with the React interface, which routes requests through the gateway to Resource_upload. Stored assets and subsection metadata supply a worker that performs transcription, document extraction, image processing, and deduplication. The resulting evidence becomes the knowledge

chunk supplied to the pedagogical prompt. Learner-profile and cognitive-load inputs arrive from other components. Two generation services return candidates, and the selector returns the chosen text with score metadata.

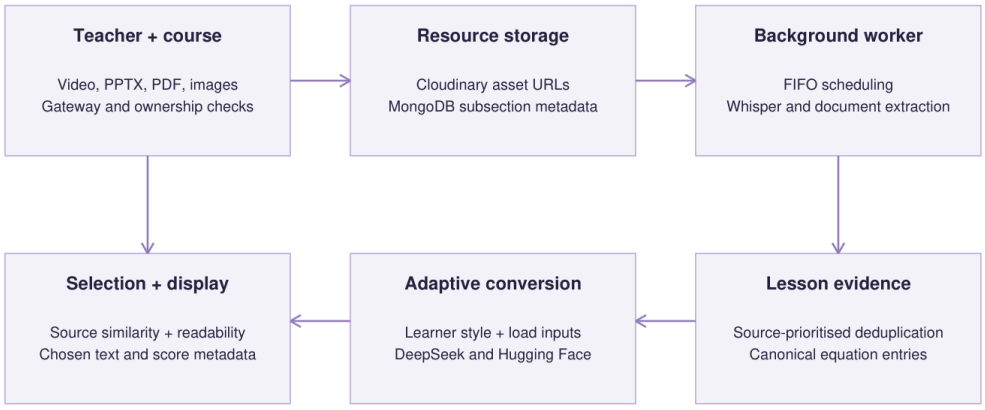



Figure 4.1. Resource processing and adaptive conversion architecture.

The architecture separates preparation from presentation transformation. Preparation is reusable across learners enrolled in a subsection; transformation is conditioned on a specific learner's state. This avoids repeating Whisper and document extraction for each learner request. It also creates a checkpoint at which resource quality could be reviewed independently of generated explanation quality. The current code provides status and extract metadata, but a strict teacher-approval checkpoint is proposed future work rather than an established requirement in the running flow.

## 4.3 Design alternatives and decisions

Synchronous preparation was rejected in favor of queued processing because ASR and extraction can be expensive. The implemented queue is intentionally simple: one global first-in, first-out (FIFO) worker in a Node.js process. It prevents that process from running two subsection preparations simultaneously and limits retained upload buffers. Its simplicity does not provide the durability, distributed locking, workload partitioning, or throughput of a dedicated message broker.

Raw concatenation was replaced with extractive deduplication so repeated material does not dominate the adaptation prompt. Presentation content is prioritised, then PDF extras,

then spoken extras. This order reflects a design preference for educator slide wording; it is not guaranteed to be optimal for every course. A generative summarizer was avoided during assembly to reduce the number of stages capable of introducing new statements. Extraction, filtering, deletion, and truncation can still alter coverage.

Equations are treated separately from prose. Approximate text similarity is unsuitable for deciding whether two expressions differing by an exponent, sign, or condition are equivalent. The implementation therefore protects detected math blocks from deletion and includes canonical expressions in the prompt. The design preserves known expressions but also preserves extraction errors when the canonical representation is wrong.

Dual generation offers alternative explanations at additional latency and cost. A weighted selector is easier to inspect than an opaque preference decision, but the scores are imperfect proxies. The particular 65% faithfulness and 35% readability weighting prioritises source consistency as a design intention. It is not a published universal weighting rule and requires local sensitivity testing.

## 4.4 Detailed design

### 4.4.1 Upload acceptance and persistence

The subsection creation controller verifies session identity, section existence, and ownership. A video is required. Videos and optional resources are uploaded to Cloudinary, and their identifiers and URLs are saved with the subsection. The controller waits for those uploads, sets knowledgeStatus to queued, schedules processing, and returns HTTP 202. Thus background processing starts after storage, while network upload itself remains within the initial request. Partial failures trigger cleanup attempts for uploaded assets.

The data model stores order, resource URLs, file names, video duration, attached images, extracted images, transcript text, deduplicated texts, equation entries, deduplication statistics, status, and status reasons. Equation records include LaTeX and source type. This supports source association, but does not provide full sentence-level provenance, content hashes, or immutable resource version identifiers.

### 4.4.2 Worker lifecycle and recovery

The worker marks a subsection processing, runs preparation, saves outcomes, and notifies the educator. A FIFO list holds waiting jobs. A map records reprocessing requests for an already active subsection; its latest request is scheduled after the current job finishes. When a waiting item is updated, it is replaced in the queue. Only the immediately available first job may retain supplied buffers; later jobs fetch persisted URLs.

At startup, queued or processing MongoDB records are rescheduled. This provides interrupted-work recovery for a single service instance, while original queue ordering is not explicitly persisted. Rebuild processing also exists through a separate setImmediate scheduler with a per-ID set. It should not be described as universally sharing the main FIFO's concurrency guarantee. Figure 4.2 distinguishes normal flow from recovery and requeue behavior.

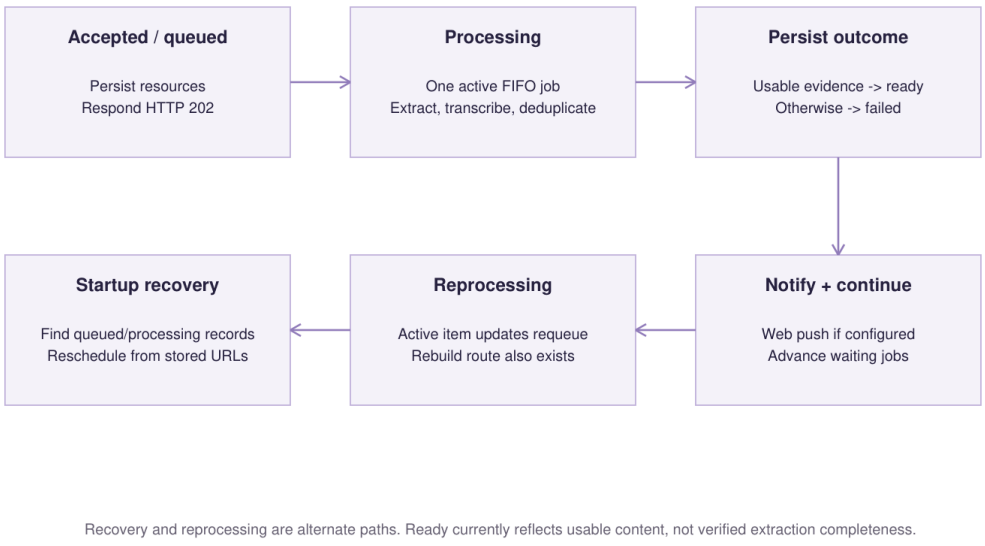



Figure 4.2. Background processing lifecycle and restart recovery.

### 4.4.3 Deduplication and source priority

Text is split into sentence-like units, with short pieces accumulated until a forty-character minimum is reached where possible. Detected dollar-delimited math blocks can be isolated and protected. Transcript units are capped at 280 through deterministic head, middle, and tail sampling. Embeddings are produced with all-MiniLM-L6-v2 when the Python dependency and model are available; otherwise term-frequency cosine

comparison is used. A Node lexical implementation supplies a further process-level fallback.

For each unit, source-specific rules determine retention. Presentation chunks are compared with kept presentation vectors. PDF chunks are compared with kept presentation and PDF vectors. Spoken chunks are compared with all document vectors and retained spoken vectors. Protected math units bypass deletion. The retained texts are joined in source order, and canonical equation entries are appended where appropriate.

### 4.4.4 Adaptation and scoring

The prompt combines course context, subsection knowledge, stored learner profile, visual–verbal category, analytic–holistic category, load, and frustration. It directs analytic explanations toward sequential steps and holistic explanations toward a big-picture overview. Visual or intermediate presentation requests include a Mermaid diagram rule. Mathematical prompts instruct verbatim copying of canonical LaTeX expressions. These are behavioral instructions to a generative service; deterministic postconditions are not enforced solely by writing them in a prompt.

Candidate comparison uses remote sentence embeddings where configured. On failure, the server computes term-frequency cosine over a shared vocabulary; despite a comment mentioning TF-IDF-style vectors, the actual fallback does not apply inverse document frequency. The browser fallback uses token-set Jaccard overlap instead. The three scoring modes have different properties and should not be pooled without recording the mode.

Table 4.2 lists current numerical settings. They are implementation constants requiring empirical validation.

Table 4.2. Configured thresholds and operational limits.

| Parameter | Current setting | Purpose |
|---|---|---|
| Video size | 100 × 1024 × 1024 bytes | Subsection upload limit |
| PPT/PDF size | 15 × 1024 × 1024 bytes each | Office document limit |
| Video duration | 900 seconds | Reject overlong lessons |
| Transcript bucket | 30 seconds in executed ASR call | Store time-associated transcript segments |
| Sentence-like minimum | 40 characters | Accumulate short split units |
| Transcript cap | 280 units | Bound deduplication workload |
| Presentation duplicate threshold | 0.70 | Within-presentation cosine comparison |

| Parameter | Current setting | Purpose |
|---|---|---|
| PDF duplicate threshold | 0.58 | Compare PDF against retained document content |
| Video/document threshold | 0.62 | Compare spoken text against document vectors |
| Video/video threshold | 0.70 | Compare against retained spoken units |
| Prompt knowledge ceiling | 12,000 characters | Bound instructional context |
| Semantic dedupe process timeout | 120 seconds | Permit lexical fallback after failure |
| Math PDF process timeout | 60 seconds | Permit text-based fallback |
| Output score weights | 0.65 similarity; 0.35 readability | Relative candidate ranking |

## 4.5 Standards and professional practice

The implementation uses HTTP status codes to distinguish accepted processing from completed results, JSON for service exchange, JWT-based identity propagation, and structured subsection relationships. These are professional design practices, not claims of certification against a formal standard. Practical quality requirements include keeping secrets outside source control, validating file content, preserving resource version information, and documenting dependencies.

For thesis reporting, IEEE-style numbered citations are used consistently. Figures and tables are numbered by chapter, and their roles are discussed in the main text. The source template's body contains eight chapters, while its contents page includes an inconsistent commercialisation chapter. This draft follows the eight-chapter body and discusses sustainability briefly in the discussion. Institutional formatting and any required commercialisation chapter should be confirmed before final submission.

# 5 IMPLEMENTATION, INTEGRATION AND TESTING

## 5.1 Development environment and technology stack

Resource_upload uses Node.js, Express, Multer, Mongoose, Cloudinary, and child-process orchestration. Python supports Whisper transcription, sentence-transformers deduplication, and PyMuPDF extraction. PPTX handling reads the Office Open XML archive and converts OMML structures through a custom JavaScript parser. Ordinary PDF text is obtained using pdf-parse. Browser conversion is integrated into the React course detail view.

The generation and selection services expose separate endpoints. The service named gpt-service hosts a Hugging Face-compatible generation workflow; the name does not imply that the candidate in this experiment is an OpenAI GPT model. Provider model IDs depend on deployment configuration. This thesis describes the code workflow and does not assume that configured providers were available during the offline verification.

## 5.2 Component implementation

### 5.2.1 Background resource preparation

processLesson.service.js implements scheduling, queue snapshots, recovery, and preparation. Processing updates the MongoDB status before expensive operations. Whisper errors are caught so that usable document content may still support a lesson. Remote office-buffer retrieval can also fail independently. The worker then extracts documents, processes images, deduplicates content, stores unique fields, and persists transcript chunks.

Preparation status is not equivalent to complete extraction quality. evaluateKnowledgeQuality identifies short extracts, small retained knowledge, and missing mathematical equations. However, the active worker uses a broader hasUsableKnowledge predicate to choose ready or failed, overriding diagnostic reasons when any usable content is present. The rebuild service follows a similar permissive readiness rule. Therefore the thesis cannot claim that all documented quality checks block publication. A non-empty transcript can allow readiness even when equation extraction is missing.

### 5.2.2 Speech transcription and noise handling

The transcription service writes a temporary media file and launches the Python Whisper script with a configured model, defaulting to base. The script calls Whisper transcription and groups returned segments into thirty-second buckets. Segments are assigned by their start time, and a bucket's end can extend when a segment crosses the nominal boundary. This preserves segment content but may create overlapping bucket intervals.

Noise filtering removes selected filler, repeated phrases, and classroom-closing material before knowledge assembly. Filtering is useful for irrelevant ASR artifacts but is not guaranteed to distinguish noise from meaningful instruction. A teacher saying a phrase that matches a filter can lose content. The workflow also cannot restore domain-specific terms that Whisper mistranscribes simply by removing filler. Transcription quality should be assessed against corrected reference transcripts.

### 5.2.3 Ordinary and mathematical resource extraction

The PPTX parser opens the archive, identifies slide XML files, sorts them numerically, and extracts text in slide order. In mathematical mode, it also locates OMML blocks and translates structures such as fractions, scripts, and n-ary operators into LaTeX. The ordinary parser does not process legacy binary PPT content. The math flag determines whether specialised paths and equation storage are enabled.

The PDF mathematical extractor uses text span geometry to infer superscript and subscript relationships and maps Unicode symbols to LaTeX commands. Heuristic tests identify math-like lines and wrap them in inline or display delimiters. If Python extraction fails, a pdf-parse and Unicode-based fallback attempts a simpler reconstruction. Neither route establishes general optical recognition of scanned formulas. The observed fixture demonstrates why successful extraction must be distinguished from faithful extraction.

Images are separately extracted from documents and stored with source metadata. The image service applies minimum-size filters and caps extracted images per source. This preserves selected visual assets, but does not establish that every diagram is extracted or correctly matched to a particular generated explanation. That association needs explicit testing.

### 5.2.4 Extractive knowledge assembly

The retained presentation, PDF, and transcript fields are joined without a generative summarizer. Canonical equations are appended as numbered entries. Course responses build the knowledge chunk from these fields and expose it to the browser adaptation flow. This means the knowledge chunk is assembled from persisted extract fields rather than necessarily stored as a separately versioned document.

The approach retains source wording at the assembly stage, subject to normalization, filtering, deduplication, and sampling. It does not support a blanket claim that retrieved content cannot contain errors. A malformed equation can be faithfully copied from an incorrect extract. Likewise, deletion can remove a qualification that makes a statement accurate. Provenance and teacher review are therefore complementary requirements.

### 5.2.5 Prompt construction and presentation adaptation

promptBuilder.js accepts known style aliases and validates load labels. Missing visual–verbal and analytic–holistic values default to Visual and Analytic. The current prompt vocabulary includes Visual, Verbal, and Intermediate, but only Analytic and Holistic for the second dimension. Thus the wider project description's moderate analytic–wholistic category is not explicitly represented in this prompt vocabulary. Invalid or unmapped inputs require careful handling rather than an assertion that every profile category is equally supported.

Knowledge is bounded at 12,000 characters. The math-preserving truncation helper attempts to close a cut display-math block by extending to a nearby closing marker, up to an additional 2,500 characters. It is not a complete parser for all mathematical syntax or concept boundaries. The selector's browser source slice is a plain 12,000-character cut, so the evidence seen during generation and scoring can differ at a math boundary. This inconsistency is a documented limitation for future alignment.

### 5.2.6 Dual conversion and source-consistency ranking

The course detail page requests Hugging Face and DeepSeek outputs through Promise.allSettled. A provider failure therefore does not immediately discard the other candidate. When outputs exist, the browser sends original subsection evidence, both candidate texts, and load level to the select-best endpoint. The server computes FRE and

readability alignment, obtains source-to-candidate similarities, and returns the selected text, model label, scores, scoring method, and warnings.

If composites tie, the implementation prefers the candidate with greater similarity; remaining ties favor the Hugging Face candidate. If only one candidate exists, the server selects it and skips faithfulness scoring. If source evidence is absent, it can rank by readability alone. If the selection endpoint is unavailable, the browser applies lexical Jaccard and FRE scoring locally. These are availability strategies, not equivalent factuality assurance. Figure 5.1 depicts the branches.

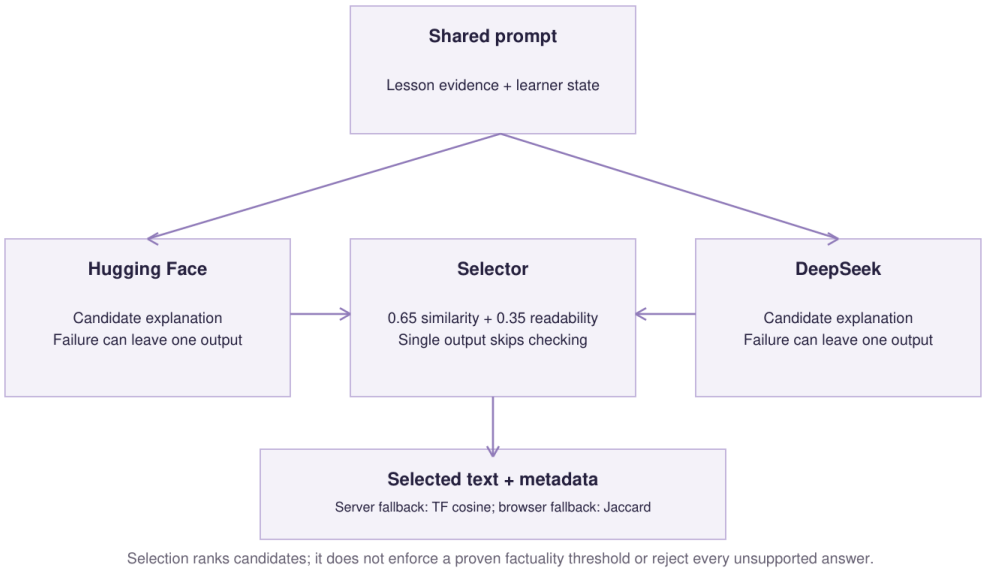



Figure 5.1. Candidate generation, selection, and degraded operation.

The selector always ranks available candidates relative to one another; it has no demonstrated minimum score at which both are rejected as insufficiently grounded. Two poor candidates can therefore yield a winner. Moreover, similar vocabulary may survive a negation, changed number, or altered formula. The implementation is accurately described as hallucination mitigation through source constraints and source-consistency ranking, not hallucination elimination.

### 5.2.7 Formula rendering and repair

The browser parses canonical equations and uses KaTeX to detect syntactically broken generated display expressions. A token-overlap heuristic can replace broken expressions with a sufficiently similar canonical entry. Syntactically valid but mathematically incorrect expressions are not necessarily replaced. The repair mechanism also inherits any

canonical extraction error. It should be assessed as a formatting-repair aid rather than a mathematical proof checker.

Further-reading generation uses a separate prompt after model selection. It permits advanced concepts and recommended resources while labeling them as supplementary. This is useful for additional study, but externally suggested links and claims require separate verification and must not be treated as teacher-source content simply because the earlier adapted lesson passed through selection.

## 5.3 Integration with the other components

The content component depends on course identity, teacher authentication, and learner enrollment flows. It receives cognitive style and load information and uses those inputs in conversion and score targeting. It does not train the cognitive classifiers. Explainable AI and recommendation services interpret learner behavior in parallel, while exam services consume teacher-provided materials for question generation. Their prediction quality and MCQ validity are outside the validation results reported here.

Table 5.1 provides code-level traceability for the individual component. It helps a reviewer locate implementation evidence without inferring behavior from interface labels.

Table 5.1. Implementation traceability.

| Concern | Principal repository path |
|---|---|
| Upload and resource persistence | Resource_upload/src/controllers/courseSubSection.controller.js |
| FIFO preparation and restart recovery | Resource_upload/src/services/processLesson.service.js |
| Startup recovery invocation | Resource_upload/server.js |
| Document format routing | Resource_upload/src/services/lessonExtract.service.js |
| PPTX/OMML conversion | Resource_upload/src/services/pptText.service.js; ommlToLatex.js |
| Math-aware PDF extraction | Resource_upload/src/python/extract_math_pdf.py |
| Transcription and timed buckets | Resource_upload/src/services/transcription.service.js; src/python/transcribe_audio.py |
| Semantic deduplication and fallback | Resource_upload/src/services/semanticDedupe.service.js; src/python/dedupe_extracts.py |
| Extractive knowledge assembly | Resource_upload/src/services/knowledgeMerge.service.js |
| Extraction diagnostics | Resource_upload/src/services/knowledgeQuality.service.js |
| Adaptive and supplementary prompts | gpt-service/components/promptBuilder.js |
| Candidate orchestration | frontend/src/pages/CourseDetail.jsx |
| Server selection and metrics | deepseek-service/lib/selectBestOutput.js; similarity.js; readability.js |

| Concern | Principal repository path |
| --- | --- |
| Browser fallback and math repair | frontend/src/utils/selectBestOutputLocal.js; assistantMath.js |

## 5.4 Verification and software testing

The focused tests intentionally avoid external services. They execute library functions using deterministic inputs and inspect outputs. Exact duplicate handling is tested with one sentence supplied across all three sources. Math protection is tested with identical dollar-delimited expressions in two document channels. Prompt checks compare visual/analytic and verbal/holistic instructions. Selector checks use fixed texts whose source relationship is known.

The PPTX fixture supplies a test of structured equation availability. The PDF fixture first tests whether text and equations are returned, then separately tests preservation of expected partial-derivative and nabla notation. This second test fails. A passing extraction-availability check is therefore retained alongside a failed correctness-related check. The initial superscript test expectation was adjusted to recognise valid grouped LaTeX, {x}^{2}; that change corrects a test expectation rather than a source-code defect.

The evidence package does not include live upload tests, interrupted-service recovery experiments, push delivery tests, API authentication tests, load tests, neural-embedding evaluation, or generated-output factuality ratings. Their mechanisms were inspected, but their runtime success is not claimed. No research application code was changed to make verification outcomes pass.

## 5.5 Deployment and operation

The resource service defaults to port 5000, the Hugging Face generation service to 5002, and the gateway to 4000 according to the repository configuration. Provider endpoints, model names, and authentication secrets are environment-dependent. Python dependencies and an FFmpeg installation are required for transcription. MongoDB and Cloudinary access are required for the upload workflow. Web push requires configured VAPID keys and a valid user subscription.

Queue snapshots expose pending and completed counts, while the frontend can display processing progress. Student visibility excludes preparing and failed statuses, and enrollment opening is computed from ready lesson availability and the absence of

preparing lessons. Failed work may be counted as completed queue work, so 100% progress is not equivalent to complete resource correctness. Multi-instance deployment would require durable job ownership, retries, concurrency control, and a consistent rebuild scheduler rather than simply duplicating the in-memory worker.

# 6 RESULTS AND VALIDATION

## 6.1 Research results

The available evidence supports technical verification findings, not causal research conclusions about student learning. The principal research observation is that pipeline stages can succeed operationally while failing to preserve instructional content. The PDF fixture is a concrete example: extraction returned equations, but an expected heat-equation symbol check failed. The result supports treating extract availability and extract fidelity as separate measures.

Another observation concerns dual-output selection. The controlled example selects the source-aligned candidate, demonstrating the implemented scoring direction under lexical fallback. This does not establish the selector's ability to identify subtle hallucinations in real model output. No statistically evaluated hallucination rate, learner-load reduction, retention gain, or dual-model improvement is reported.

## 6.2 Product and technical results

The JavaScript checks passed thirteen of thirteen final assertions. Python checks passed five of six. Across both sets, eighteen of nineteen focused checks passed. This count is a descriptive verification summary; it is not a reliability estimate or an accuracy score for the complete platform. Table 6.1 records their scope.

Table 6.1. Offline verification results.

| Check group | Observed result | Interpretation |
|---|---|---|
| OMML elementary conversion, J01–J02 | 2/2 passed | Fraction and grouped superscript cases return expected LaTeX |
| Presentation fixture and legacy format, J03–J04 | 2/2 passed | PPTX fixture returns five equations; legacy PPT returns empty text |
| Knowledge ordering and diagnostic warning, J05–J06 | 2/2 passed | Source order and canonical entry retained; missing equations diagnosed |
| Prompt and readability rules, J07–J09 | 3/3 passed | Requested layouts and configured FRE targets are present |
| Similarity and selection branches, J10–J13 | 4/4 passed | Lexical fallback, dual ranking, single output, and missing output behavior verified |
| Duplicate and math protection, P01–P02 | 2/2 passed | One exact duplicate retained; protected repeated math remains |

| Check group | Observed result | Interpretation |
|---|---|---|
| Transcript boundaries and sampling, P03/P05 | 2/2 passed | Bucket spillover preserved; 400 units reduced to 280 |
| PDF extract availability, P04 | Passed | Returns 1,147 characters and eight equation candidates |
| PDF heat-equation symbol fidelity, P06 | Failed | Expected partial and nabla commands not preserved |

The PPTX extractor returned 1,102 characters and five equation strings for the existing heat-equation presentation. The expressions include Fourier's law, the heat equation, a Gaussian expression, a spatial eigenvalue equation, and a series expression. Counting these entries demonstrates availability; mathematical equivalence of every extracted expression has not been independently rated.

The PDF mathematical path returned eight candidates, including strings containing substituted multiplication-dot commands where mathematical operators or symbols were expected. One prose-like boundary-condition line was also included as an equation candidate. The targeted assertion checked whether a recovered expression preserved both partial and nabla commands; it failed. The precise cause of character loss must be isolated between fixture encoding and parser reconstruction before attributing it solely to the parser. Either way, the workflow did not preserve the intended fixture notation in this execution.

The exact-duplicate fixture retained one of three repeated source chunks, corresponding to a two-thirds reduction within that artificial example. It does not estimate real lecture redundancy reduction. With math protection enabled, two identical expressions were both retained. This confirms that the preservation policy can intentionally trade compression for caution.

In the two-candidate selection fixture, the exact source-aligned candidate received lexical similarity 1.0000, FRE 102.05, readability match 0.2590, and composite 0.7407. The unrelated candidate received similarity 0.0000, FRE 54.73, readability match 0.7946, and composite 0.2781. The source-aligned candidate won despite poorer alignment with the High-load target of 65. This illustrates the relative importance of the configured similarity weight. It is a scorer test using authored text, not a comparison of actual Hugging Face and DeepSeek generation quality.

## 6.3 User or stakeholder validation

No teacher interviews, usability questionnaire, learner trial, or independent content ratings were available. Therefore the thesis does not claim stakeholder acceptance. A proposed teacher-validation activity would ask educators to assess source coverage, equation correctness, section coherence, and the usefulness of processing diagnostics. A proposed learner activity would compare explanations under controlled lesson and prior-knowledge conditions, while allowing learners to choose a preferred presentation rather than treating assigned style categories as fixed traits.

These activities require approved instruments, consent procedures, recruitment records, and documented analysis before their results can be included. Their planned inclusion should not be confused with completed validation.

## 6.4 Integration results

Static tracing identifies an implemented path from subsection upload to queued processing, persisted extracts, course response knowledge, prompt construction, concurrent candidate requests, and selection metadata. The local modules can be loaded and exercised for the reported helper behaviors. This establishes interface and code-path evidence, but does not confirm end-to-end service operation against live MongoDB, Cloudinary, generation providers, and browser sessions.

Two integration concerns were identified. First, generation's math-aware truncation and scoring's plain character slice can produce slightly different source contexts. Second, helper-level quality warnings can be bypassed by the worker's usable-content readiness rule. Both should be checked in live integration tests because they affect what students receive even if individual functions behave as implemented.

## 6.5 Summary of findings

The current artefact implements resource organisation, asynchronous preparation, specialised extraction, extractive assembly, profile-conditioned prompting, and dual-candidate ranking. Focused verification supports selected helper behavior and reveals a failure in a mathematical fixture. The findings justify continued development and controlled evaluation, while preventing claims of complete fidelity, hallucination elimination, or proven educational effectiveness.

# 7 DISCUSSION

## 7.1 Interpretation of findings

The pipeline's strongest design feature is separation of educator evidence from learner-specific presentation. Preparation produces reusable subsection knowledge, and adaptation changes how that knowledge is explained. This separation permits stage-level evaluation and avoids using model-generated summaries as the only evidence supplied to another generator. Nevertheless, extractive assembly is only as dependable as its inputs and retention policy.

The PDF result shows the danger of labeling extracted equations canonical without verifying them. Canonical means that later stages are instructed to preserve a stored representation; it does not mean the representation was recovered correctly. The presentation path's five returned formulas and the PDF path's symbol failure also show why heterogeneous inputs need separate evaluation. One successful format cannot validate another.

## 7.2 Comparison with literature, baselines and existing solutions

The component aligns with literature motivations for source-aware generation [1], semantic comparison [2]–[4], structured equation handling [5]–[8], and focused learning units [9], [10]. Its evidence is much narrower than those motivations. It has not demonstrated that its thresholds outperform exact matching, that its equation extractor matches LATTE, or that its short lessons produce the learner benefits reported by other systems.

The supplied concept-explanation paper [16] cannot validate the project's FRE formula because it evaluates a different kind of readability. Similarly, general LLM evaluation guidance [17] does not justify the precise selection weights as a measured optimum. Appropriate baselines for this component are its own raw-concatenation, lexical-deduplication, single-candidate, and readability-only variants under fixed lesson materials and provider settings.

## 7.3 Research contribution

The proposed research contribution is a framework for analysing the reliability chain between upload and adaptation. It treats equation fidelity, evidence retention, learner-conditioned presentation, and output selection as connected but independently testable responsibilities. The documented fallback modes and permissive readiness rule are especially relevant because a production interface can otherwise hide the difference between available output and verified output.

This contribution remains an artefact-based proposition supported by focused technical checks. Establishing generalizable superiority requires a broader benchmark and user validation. The thesis should not present integration alone as proof of pedagogical novelty.

## 7.4 Product contribution

The implemented product contribution is a teacher-facing upload workflow with background preparation and a learner-facing conversion path that can compare two alternatives. Source ordering, math-aware extraction, canonical equation entries, and score metadata make the process more inspectable than an unexplained single response. Recovery from persisted queued states and fallback scoring also support continuity when dependencies fail.

However, the product currently favors availability in several branches. A single provider result is selected without server faithfulness checking, and a non-empty transcript can allow a lesson to become ready despite extraction warnings. These are defensible prototype choices only when their limitations are explicit. A deployment intended for assessed learning should prioritize verified source integrity over the appearance of uninterrupted output.

## 7.5 Limitations and threats to validity

Construct validity is limited because cosine similarity and lexical overlap do not measure factual entailment, and FRE does not measure cognitive load. Technical word counts and syllable heuristics can also behave poorly on symbols, code, abbreviations, and equations. Readability targeting penalises deviations in both directions, so text easier than the selected target can receive a lower match; that behavior needs pedagogical justification.

Internal validity is limited by authored fixtures and lack of human blind ratings. Model candidates were not generated during the offline tests. The test suite deliberately forces lexical fallback and therefore provides no evidence about neural embeddings or provider-generation quality. External validity is limited by one presentation and one PDF fixture. Short cases cannot represent scanned documents, diverse equations, accents, diagrams, and long-form lectures.

Operational limitations include an in-memory global queue, potential multi-instance duplication, a separately scheduled rebuild route, lack of a demonstrated Whisper timeout, and dependency on external asset availability. Queue completion does not certify extraction correctness. Content limitations include lossy transcript sampling, context truncation, deletion of useful paraphrases, incomplete sentence-level provenance, and preservation of incorrect canonical expressions. Prompt vocabulary and default labels can misrepresent missing or moderate learner states.

Bibliographic validity also matters. Some spreadsheet claims make direct methodological connections that the cited studies do not establish. A reference audit is therefore included rather than silently retaining those interpretations. The review is targeted, and final submission should strengthen competing-system coverage and verify reference versions in accordance with institutional requirements.

## 7.6 Implications

For educators, the component could reduce repetitive preparation and provide a starting point for learner-specific explanations, but teacher verification remains essential for mathematical resources and high-stakes course content. For developers, reliability should be tracked at each stage with resource hashes, extraction versions, source spans, candidate records, and explicit degradation indicators. For researchers, content coverage and correctness must be measured alongside readability and operational success.

A sustainable deployment should monitor transcription compute, duplicate model requests, asset storage, and recurring provider costs. Caching prepared subsection knowledge helps avoid repeated extraction. Any commercial offering would require institution-specific data agreements, content rights management, accessible presentation, and validated utility; no market-size, revenue, or licensing claim is made here. The component's educational intent is consistent with supporting access to quality learning,

but an actual contribution to SDG 4 would require evidence of adoption and outcomes rather than a feature description.

# 8 CONCLUSION AND FUTURE DIRECTIONS

## 8.1 Conclusion

This thesis documents Lumora's resource upload and source-grounded adaptive content conversion component. The implemented pipeline organises teacher resources, schedules background preparation, extracts speech and document content, treats equations through specialised paths, reduces repeated evidence, and generates two candidate learning aids for inspectable selection. It connects the content infrastructure to learner cognitive style and load inputs supplied by other components.

Focused verification confirms selected deterministic behaviors and exposes incomplete mathematical preservation in a PDF fixture. The similarity-based checker supports ranking but does not prove the absence of hallucination. Consequently, the defensible conclusion is that the component provides an implemented and testable integration architecture with known limitations. Evidence of learning improvement, robust mathematical fidelity, and reduced hallucination across real lessons remains to be collected.

## 8.2 Achievement of objectives

Table 8.1 relates objectives to evidence. Implementation and verification are assessed separately.

Table 8.1. Objective achievement and remaining validation.

| Objective | Current evidence | Remaining work |
|---|---|---|
| O1: Resource organisation and upload | Controllers, ownership checks, limits, and schemas inspected | Live upload/security tests |
| O2: Asynchronous preparation | FIFO, status updates, snapshots, and restart scan inspected | Recovery and load experiments |
| O3: Multiformat and math processing | PPTX/PDF helper outputs observed | Correct PDF symbol loss; broaden formula benchmark |
| O4: Redundancy reduction | Exact duplicate and protected math tests observed | Measure unique information retention and threshold sensitivity |
| O5: Learner-conditioned conversion | Prompt-layout tests and integration path observed | Evaluate generated layouts and learning usefulness |
| O6: Dual-output ranking | Composite calculation and fallback branches tested | Calibrate factuality metric, weights, and rejection rules |

## 8.3 Individual research and product contributions

The individual component contributes the upload-to-knowledge and knowledge-to-adaptation connection within Lumora. It supplies resource processing mechanisms, mathematical representation handling, source-preserving assembly, and candidate-level selection metadata. The present thesis adds transparent technical evidence and identifies where implementation comments, interface promises, and actual conditions differ.

Personal contribution descriptions are based on the role supplied by the candidate. Team records should confirm the scope before signatures are added. Neighboring cognitive classifiers, LIME/SHAP analytics, and exam generation are described for context without assigning their implementation to this candidate.

## 8.4 Future directions

First, introduce a teacher-review checkpoint for extracted knowledge and equations. Separate preparation completion from verified readiness, retain diagnostic warnings, and block mathematically flagged lessons when required expressions are absent or corrupted. Add formula rendering comparisons, syntax validation, and semantic equivalence review before treating extracts as canonical.

Second, move processing to a durable queue with explicit leases, bounded concurrency, retries, and observable failure states. Unify or coordinate rebuild work with preparation scheduling, add transcription timeouts, and test restart behavior across multiple service instances. Record immutable resource versions so a reprocessed lesson can be traced to the exact uploaded files.

Third, evaluate deduplication using teacher-labeled duplicate pairs and distinct concept inventories. Calibrate thresholds independently for neural and lexical representations. Replace blind transcript sampling and plain character cuts with concept-aware evidence selection while preserving conditions, dependencies, and equations.

Fourth, supplement similarity ranking with claim-level support checking and a reject-or-review outcome. Validate numeric values, negation, equation identity, and source citations. Record provider model revisions and assess both single-candidate and degraded modes explicitly. Evaluate 0.65/0.35 weighting against human judgments rather than assuming it is optimal.

Finally, run controlled teacher and learner studies after technical fidelity improves. Measure source coverage, content usefulness, quiz performance, delayed retention, and self-reported effort. Provide learners with presentation choices and treat detected styles as revisable input signals. These steps would convert a promising integration artefact into a more rigorously validated educational contribution.

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# GLOSSARY

Canonical equation: A stored extracted expression used as the reference for later copying or formatting repair. Its designation does not imply that extraction has been mathematically verified.

Cognitive-load-aware conversion: Adaptation conditioned on a load estimate supplied to the component. It does not itself establish that the resulting text reduces measured cognitive load.

Dual conversion: Generation of two alternative transformations of the same instructional input, followed by relative scoring.

Extractive knowledge assembly: Combining retained source extracts without generating a new prose summary at the assembly stage.

Faithfulness proxy: A computable approximation of source consistency, such as cosine similarity. A proxy can fail to detect unsupported statements or contradictions.

Graceful degradation: Continuing with a reduced capability when a preferred dependency fails, while recording that the method has changed.

Microlearning: Organisation of learning around focused, manageable instructional units; not equivalent to arbitrary short technical chunks.

Near duplicate: Text similar enough to previously represented content to be considered redundant under a specified criterion.

Readability match: Closeness of a candidate's FRE score to a configured load-dependent target, not a direct measure of learning effort.

# APPENDICES

## Appendix A. Research artefacts and reference audit

The accompanying verify_component.js and verify_component.py scripts reproduce the focused checks described in Chapters 3 and 6. Their outputs are verification_node.json and verification_python.json. Run from the repository with the already installed relevant dependencies. The Python script exits with a failure status for the PDF fidelity assertion; that failure is retained as evidence rather than suppressed.

Commands: node thesis/content_conversion/verify_component.js; python thesis/content_conversion/verify_component.py.

The test suite does not start servers, access databases, use credentials, or request model-generated content. The JavaScript script clears the embedding token in its child process to force the recorded fallback. The Python suite replaces the neural embedding helper with a deliberate exception, ensuring that the deduplication fixture measures lexical behavior. The tests are supplementary verification prepared during thesis drafting and should be attributed as such.

Table A.1 maps every populated spreadsheet reference to its treatment in the thesis. Verification uses accessible primary records or author material; it does not imply that every full paper was exhaustively read.

Table A.1. Spreadsheet reference audit.

| Spreadsheet reference | Thesis reference | Audit outcome |
|---|---|---|
| JATS-Con Proceedings 2016 | [5] | Corrected to the specific Gebhard and Rosenblum chapter; supports specialised math conversion, not automatic fidelity |
| Benchmarking Document Parsers, 2025 | [7] | Title/authors verified; version updated in 2026; motivates meaning-sensitive formula evaluation |
| LATTE / AAAI 2025 | [6] | Title and acceptance verified; related technique, not installed extraction subsystem |
| Spoken expressions to LaTeX, 2024 | [8] | Corrected to MathBridge; not proof of delimiter sufficiency or implemented spoken-math conversion |

| Spreadsheet reference | Thesis reference | Audit outcome |
|---|---|---|
| Sentence-BERT | [2] | Foundational semantic representation rationale |
| MiniLM | [3] | Distillation rationale; no claim that this component reproduces paper benchmarks |
| RETSim | [4] | Relevant alternative; not the project's implemented model |
| The Reel Deal | [10] | Own study verified in abstract; no imported Lumora outcomes |
| ScienceDirect microlearning review | [9] | Corrected complete title/authors and online/issue dates; duration claims not treated as universal rules |
| Partial Mobilization | [13] | Different domain; does not validate local 0.62/0.70 cutoffs |
| NovAScore | [14] | Novelty evaluation context; numeric thresholds not adopted as verified educational settings |
| Maynez et al. | [1] | Supports concern about abstractive faithfulness; extraction can still contain errors |
| Clinical chunking comparison | [15] | Domain-qualified support; publisher access was limited, indexed primary record consulted |
| Concept-based explanation readability | [16] | Different constructs from FRE and lecture rewriting; direct metric-equivalence claim rejected |
| Databricks evaluation guidance | [17] | Technical article; no verification of the precise 65/35 weighting |
| ExpertPrompting | [18] | Structured role prompting motivation; no guaranteed transfer of reported results |
| Cognitive load theory | [11] | Foundational rationale, not validation of configured FRE targets |
| Engineering learning styles | [12] | Author-hosted reprint used instead of Drive link; no claim that matching styles proves learning improvement |

Additional completion requirements are the group ID, candidate confirmation of personal contribution, any institutional formatting requirements, actual learner or teacher study evidence if conducted, original live integration logs, and final bibliographic verification. The final report should retain truthful limitations or replace them only with subsequently collected evidence.

## Appendix B. Academic integrity and AI-use disclosure

Tool: OpenAI Codex assistant, used for draft preparation, repository tracing, literature-record checks, deterministic verification script creation, and document generation. Python with openpyxl and PyMuPDF was used to inspect the supplied spreadsheet and PDF and to generate the review copy. Node.js executed component helper checks. The declaration and supervisor certification remain unsigned.

Purpose and affected sections: AI assistance structured and drafted the full thesis, including the abstract, chapters, references, glossary, and appendices, from the supplied material and inspected implementation. It also created the accompanying verification scripts and document-export script. The research application itself was not modified as part of thesis preparation.

Representative workflow: “Draft an individual thesis for the resource upload and content conversion component using the supplied thesis PDF and evidence spreadsheet; inspect the repository, distinguish implemented behavior from proposed evaluation, and do not invent study results.” Follow-up verification traced background preparation, math paths, deduplication thresholds, prompt constraints, dual-candidate scoring, and fallback branches.

Verification performed: Reference titles, authors, and scope were checked against accessible primary sources; implementation statements were traced to files; nineteen focused assertions were executed; a failed PDF fidelity assertion was retained; generated document structure and citation numbering were checked. These activities do not replace a complete human literature review, proof of personal code authorship, or supervisor review.

Candidate review required: Confirm that this disclosure satisfies module policy, check every claim and reference, verify the mathematical examples, review all attribution, provide missing administrative information, and approve the complete thesis before signing. No statement is made that AI use has already been approved by the institution.